

PVsyst - Simulation report

Grid-Connected System

Project: **Battery (2)**

Variant: **8.82**

No 3D scene defined, no shadings

System power: **8.82** kWp

Kampung Chuang - Malaysia

Author

NORTHERN SOLAR SDN. BHD. (Malaysia)



Project: Battery (2)

Variant: 8.82

PVsyst V8.1.3

VC0, Simulation date:
with V8.1.3

NORTHERN SOLAR SDN. BHD. (Malaysia)

Project summary

Geographical Site

Kampung Chuang

Malaysia

Situation

Latitude 3.4869 °(N)

Longitude 101.6284 °(E)

Altitude 22 m

Time zone UTC+8

Project settings

Albedo 0.20

Weather data

Kampung Chuang

Meteonorm 9.0 dll, Sat=100% - Synthetic

System summary

Grid-Connected System

Simulation for year no 1

No 3D scene defined, no shadings

Orientation #1

Fixed plane

Tilt/Azimuth °

Orientation #2

Fixed plane

Tilt/Azimuth °

Near Shadings

no Shadings

System information

PV Array

Nb. of modules units

Pnom total kWp

Inverters

Nb. of units unit

Total power kWac

Pnom ratio

Battery pack

Storage strategy: Self-consumption

Nb. of units unit

Voltage 51 V

Capacity Ah

User's needs

Monthly values

Results summary

Simulation step Hourly

Produced Energy kWh/year

Used Energy kWh/year

Specific production kWh/kWp/year

Perf. Ratio PR %

Solar Fraction SF %

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Project: [REDACTED]

Variant: [REDACTED]

PVsyst V8.1.3VC0, Simulation date:
with V8.1.3

NORTHERN SOLAR SDN. BHD. (Malaysia)

General parameters**Grid-Connected System**

No 3D scene defined, no shadings

Simulation step

Hourly

Orientation #1**Fixed plane**

Tilt/Azimuth [REDACTED] °

Orientation #2**Fixed plane**

Tilt/Azimuth [REDACTED] °

Models usedTransposition Perez
Diffuse Perez, Meteonorm
Circumsolar separate**Horizon**

Average Height 2.6 °

Near Shadings

no Shadings

User's needs

Monthly values

Storage

Kind Self-consumption

Jan.	Feb.	Mar.	Apr.	May	June	July	Aug.	Sep.	Oct.	Nov.	Dec.	Year	
[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	KWh/mth

PV Array Characteristics**PV module**

Manufacturer Astronergy

Model CHSM66RN(DG)/F-BH-630

(Custom parameters definition)

CHSM66RN(DG)F-BH-630.PAN

Unit Nom. Power 630 Wp

Inverter

Manufacturer

Model Hoymiles

(Custom parameters definition)

Unit Nom. Power 5.00 kWac

Array #1 - PV Array

Orientation 1

Tilt/Azimuth 20/90 °

Number of PV modules 7 units

Nominal (STC) 4410 Wp

Modules 1 unit x 7 In series

At operating cond. (50°C)

Pmpp 4115 Wp

U mpp 264 V

I mpp 16 A

Number of inverters 1 * MPPT 50% 0.5 unit

Total power 2.5 kWac

Operating voltage 90-460 V

Pnom ratio (DC:AC) 1.76

Array #2 - Sub-array #2

Orientation 2

Tilt/Azimuth 20/-90 °

Number of PV modules 7 units

Nominal (STC) 4410 Wp

Modules 1 unit x 7 In series

At operating cond. (50°C)

Pmpp 4115 Wp

U mpp 264 V

I mpp 16 A

Number of inverters 1 * MPPT 50% 0.5 unit

Total power 2.5 kWac

Operating voltage 90-460 V

Pnom ratio (DC:AC) 1.76

Total PV power

Nominal (STC) 9 kWp

Total 14 modules

Module area 37.8 m²

Cell area 35.3 m²

Total inverter power

Total power 5 kWac

Number of inverters 1 unit

Pnom ratio 1.76

No power sharing

**Grid storage - Self-consumption****Battery**

Manufacturer

Hoymiles

Model

LB-16D-G3**Battery pack**

Nb. of units

1 unit

Voltage

51 V

Nominal Capacity (C10)

314 Ah

Nominal energy

16.1 kWh

Operating SOC limits

95.0 / 20.0 %

Useable energy

12.1 kWh

Operating temperature

Fixed 20 °C

Battery input charger

Model

Generic

Max. charging power

7.50 kWdc

Max./Euro effic.

98.0 / 96.5 %

Battery to Grid inverter

Model

Generic

Max. discharging power

1.40 kWac

Max./Euro effic.

98.0 / 96.5 %

Array losses**Array Soiling Losses**

Loss Fraction

1.5 %

Thermal Loss factor

Module temperature according to irradiance

Uc (const)

20.0 W/m²K

Uv (wind)

0.0 W/m²K/m/s

LID - Light Induced Degradation

Loss Fraction

1.45 %

Module Quality Loss

Loss Fraction

-0.75 %

Module mismatch losses

Loss Fraction

1.00 % at MPP

Module average degradation

Year no

1

Loss factor

0.55 %/year

Imp / Vmp contributions

80% / 20%

Mismatch due to degradation

Imp RMS dispersion

0.4 %/year

Vmp RMS dispersion

0.4 %/year

IAM loss factor

ASHRAE Param.: IAM = 1 - bo (1/cosi - 1)

bo Param.

0.05

DC wiring losses

Global wiring resistance

10 mΩ

Loss Fraction

2.00 % at STC**Array #1 - PV Array**

Global array res.

550 mΩ

Loss Fraction

3.00 % at STC**Array #2 - Sub-array #2**

Global array res.

275 mΩ

Loss Fraction

1.50 % at STC**System losses****Unavailability of the system**

Time fraction

1.0 %

3.7 days,

3 periods

AC wiring losses**Inv. output line up to injection point**

Inverter voltage

220 Vac mono

Loss Fraction

1.00 % at STC

Global System

Wire section

Copper **2 x 6** mm²

Wires length

9 m



Main results

Simulation step

Hourly

System Production

Produced Energy 11221 kWh/year

Used Energy 11390 kWh/year

Specific production 1272 kWh/kWp/year

Perf. Ratio PR 73.73 %

Solar Fraction SF 75.78 %

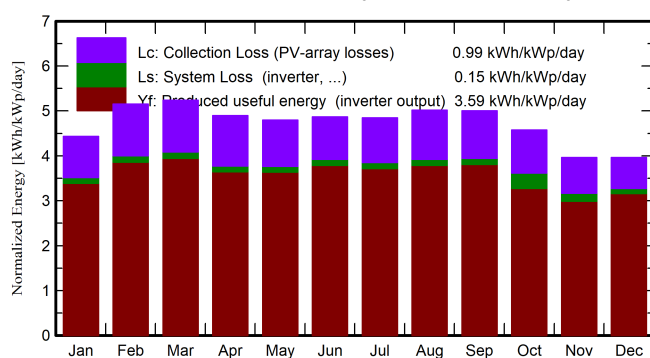
Battery ageing (State of Wear)

Cycles SOW 68.8 %

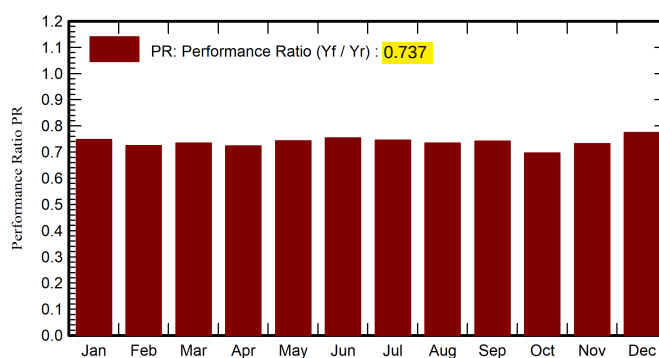
Static SOW 90.0 %

Battery lifetime 3.2 years

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor kWh/m ²	DiffHor kWh/m ²	T_Amb °C	GlobInc kWh/m ²	GlobEff kWh/m ²	EArray kWh	E_User kWh	E_Solar kWh	E_Grid kWh	EFrGrid kWh
January										
February										
March										
April										
May										
June										
July										
August										
September										
October										
November										
December										
Year										

Legends

GlobHor Global horizontal irradiation

DiffHor Horizontal diffuse irradiation

T_Amb Ambient Temperature

GlobInc Global incident in coll. plane

GlobEff Effective Global, corr. for IAM and shadings

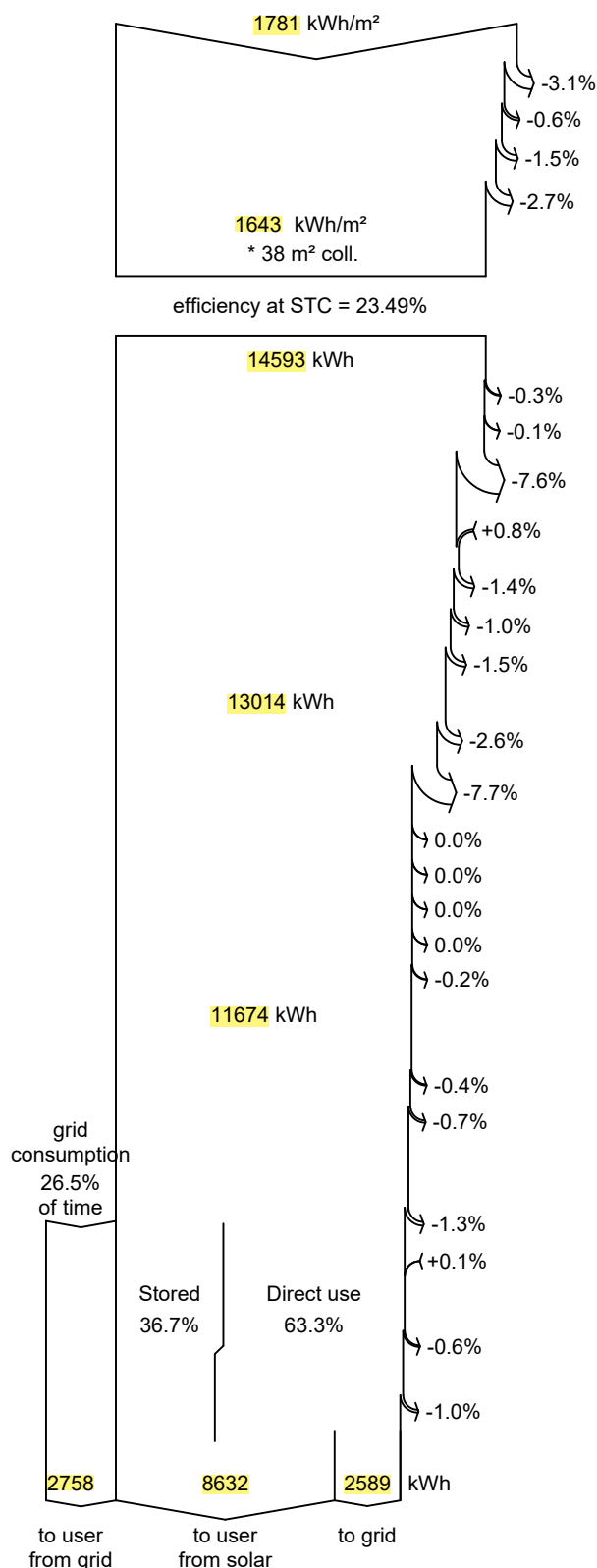
EArray Effective energy at the output of the array

E_User Energy supplied to the user

E_Solar Energy from the sun

E_Grid Energy injected into grid

EFrGrid Energy from the grid

**Loss diagram****Global horizontal irradiation****Global incident in coll. plane**

Far Shadings / Horizon

Soiling loss factor

IAM factor on global

Effective irradiation on collectors

PV conversion

Array nominal energy (at STC effic.)

Module Degradation Loss (for year #1)

PV loss due to irradiance level

PV loss due to temperature

Module quality loss

LID - Light induced degradation

Module array mismatch loss

Ohmic wiring loss

Array virtual energy at MPP

Inverter Loss during operation (efficiency)

Inverter Loss over nominal inv. power

Inverter Loss due to max. input current

Inverter Loss over nominal inv. voltage

Inverter Loss due to power threshold

Inverter Loss due to voltage threshold

Night consumption

Available Energy at Inverter Output

AC ohmic loss

System unavailability

Battery IN, charger loss

Battery Stored Energy balance

Battery StorageBattery global loss
(1.6% of the battery contribution)

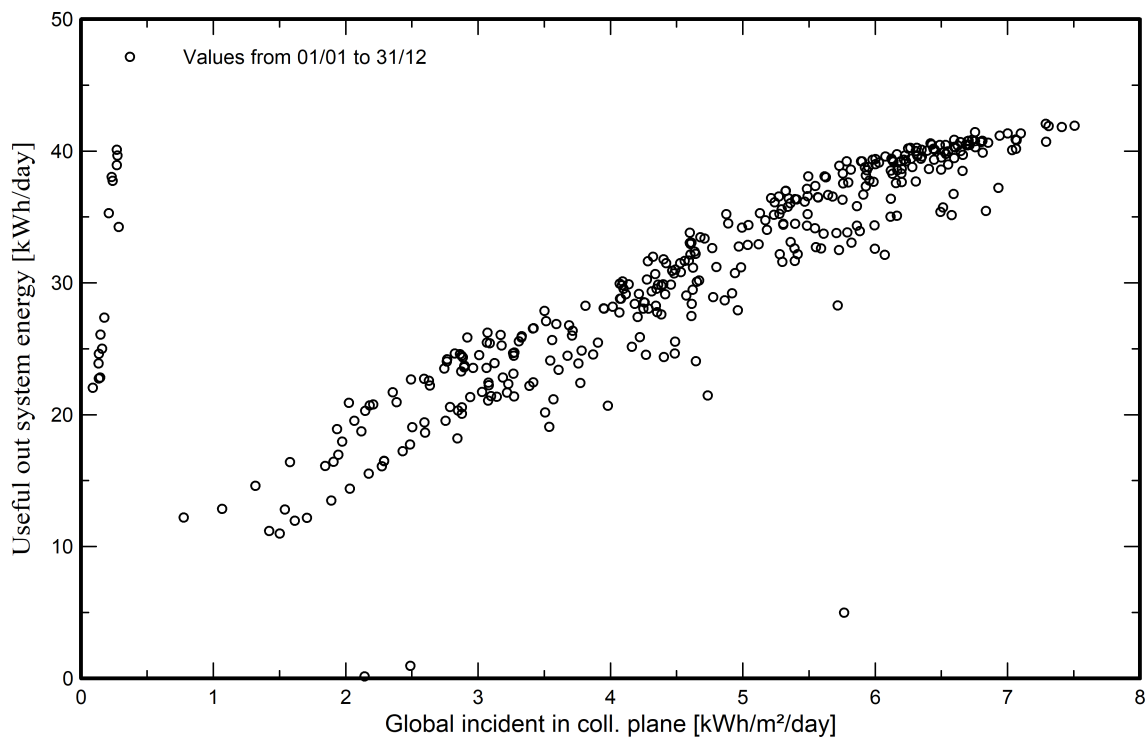
Battery OUT, inverter loss

Dispatch: user and grid reinjection

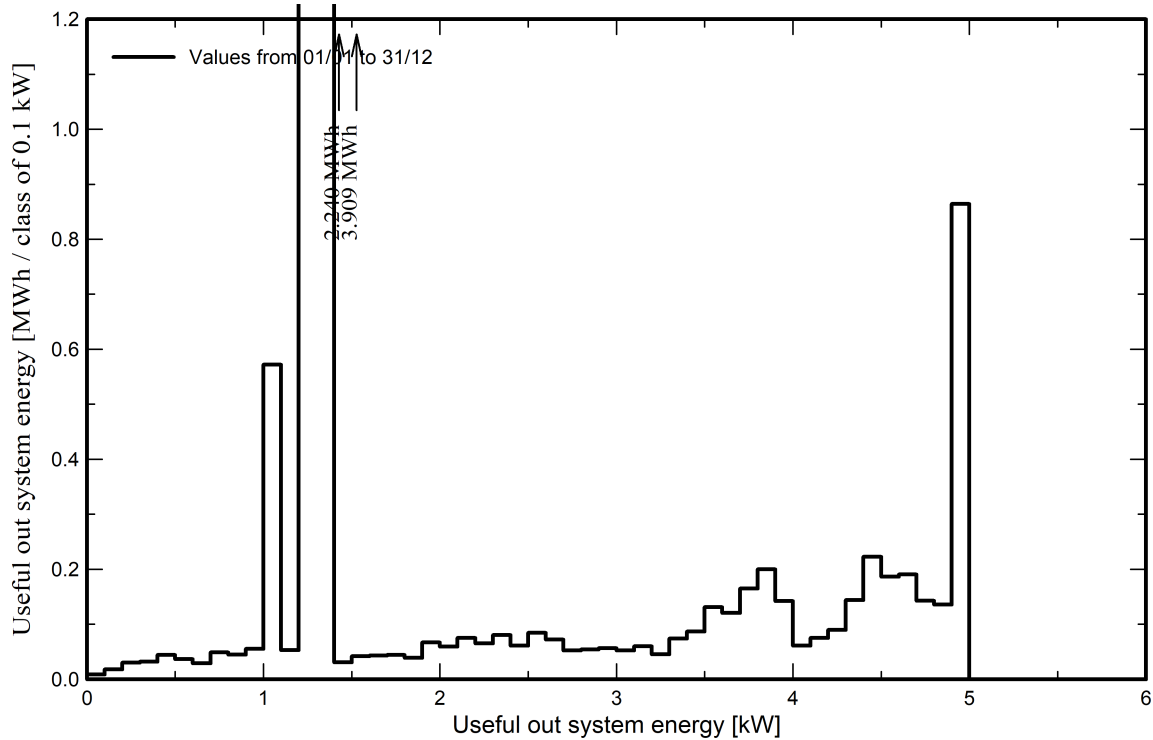


Predef. graphs

Daily Input/Output diagram



System Output Power Distribution





P50 - P90 evaluation

Weather data

Source Meteonorm 9.0 dll, Sat=100%
Kind Specific year
Year Synthetic
Year-to-year variability(Variance) 4.0 %

Specified Deviation

Year deviation from average 0.0 %

Global variability (weather data + system)

Variability (Quadratic sum) 5.9 %

Simulation and parameters uncertainties

PV module modelling/parameters 1.0 %
Inverter efficiency uncertainty 0.5 %
Soiling and mismatch uncertainties 1.0 %
Degradation uncertainty 1.0 %
Custom variability 4.0 %

Annual production probability

Variability 666 kWh
P50 11221 kWh
P90 10367 kWh
P75 10772 kWh

Probability distribution

